

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY,
BELAGAVI - 590 014, KARNATAKA**



SYNOPSIS ON

“Parkinson’s disease detection using Machine Learning”

**A Major project synopsis submitted in partial fulfilment of requirement for the 6th
semester B.E Course during the academic year 2024-2025**

Submitted by

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Under the guidance of

DR. Margesh Keskar



DEPARTMENT OF CSE (DATA SCIENCE) ENGINEERING

GURU NANAK DEV ENGINEERING COLLEGE

BIDAR-585 403, KARNATAKA

ACADEMIC YEAR 2025-2026

To,

DATE: -22/04/2025

The Head of Department

CSE (Data Science) DEPARTMENT

GNDEC

BIDAR

Subject: - “Regarding the Permission for carrying out the final year project”

Respected Sir,

We hereby requested you to grant the permission to carry out our project work on “Parkinson’s Disease Detection Using Machine Learning” under the guidance of “Dr. Margesh keskar”. So, we request you to approve our topic and have your kind consideration into the matter.

Thanking You

Biradar Vaishnavi (3GN22CD017)

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Dr. Margesh keskar

(Project Guide)

Prof. Syed Saqlain Ahmed

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Dr. Margesh keskar

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ABSTRACT

Parkinson's Disease (PD) is a chronic and progressive neurodegenerative disorder that primarily affects the motor functions of the human body due to the degeneration of dopamine-producing neurons in the brain. It manifests through symptoms such as tremors, muscular rigidity, bradykinesia (slowness of movement), and postural instability, which gradually worsen over time and significantly impact a patient's quality of life. Early detection of Parkinson's Disease plays a crucial role in controlling symptom progression and improving treatment effectiveness. However, conventional diagnostic approaches are largely based on clinical observation and expert judgment, which may lead to delayed diagnosis and possible inaccuracies, especially in the early stages of the disease.

This project aims to develop an intelligent and automated system for the early prediction of Parkinson's Disease using advanced Machine Learning (ML) techniques. The system utilizes a biomedical voice dataset obtained from the UCI Machine Learning Repository, which consists of 195 instances and 23 attributes. These attributes include critical vocal features such as jitter, shimmer, fundamental frequency, and harmonics-to-noise ratio, which are highly indicative of voice impairments associated with Parkinson's Disease. Since speech disorders are among the early symptoms of PD, analyzing voice measurements provides a non-invasive and efficient way for early detection.

The proposed methodology involves multiple stages including data preprocessing, feature engineering, model development, and performance evaluation. In the preprocessing phase, the dataset is cleaned, normalized, and scaled to enhance data quality and remove inconsistencies. Feature selection techniques are applied to identify the most relevant attributes contributing to accurate prediction. Machine learning algorithms such as Random Forest and XGBoost are then implemented to build classification models. These models are trained using labeled data to classify individuals as either healthy or affected by Parkinson's Disease. The performance of the models is evaluated using standard metrics such as accuracy, precision, recall, and confusion matrix to ensure robustness and reliability.

Experimental results show that XGBoost outperforms Random Forest, achieving about 94.8% accuracy compared to 92%. The system provides a reliable, cost-effective, and non-invasive method for early detection of Parkinson's Disease. It can assist healthcare professionals in diagnosis and has potential for real-time applications, with future scope in deep learning and mobile-based systems.

